

Data acquisition and cleaning:

Data sources

The collisions data is recorded by Seattle Department of Transportation, Traffic Management Division, Traffic Records Group and maintained by ArcGIS and can be found in the Kaggle dataset [here](#) and the metadata can be found from [here](#). The collisions dataset includes all types of collisions from year 2004 to present.

Data Cleaning

The data downloaded from the above-mentioned source was saved in a table. There were many entries in the dataset which said that no person, pedestrian, cyclists or vehicles were involved in those collisions. Analyzing such entries, a bit further they were found to be irrelevant and hence deleted from the dataset.

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There were several columns (features) in the dataset that were added by various authorities to mainly identify the incident instance and were irrelevant to our final goal of developing a predictive model. Those columns were dropped. Our aim is to build a predictive model which can foretell or predict accurately the severity of any future accident given the conditions (features) that can be pre specified. There were several columns in our dataset which only made sense only when a collision had already taken place; so those columns were also dropped. Also we want our model to be independent of the exact locations of the collisions, therefore columns indicating the exact locations of the collisions were dropped and rather the columns describing the properties of such locations were kept. Below is a table of the dropped columns and the reason for their dropping:

There were a lot of missing values as well as data inconsistencies in many of the kept features fields. For example, the „ADDRTYPE“ feature which is a categorical variable, describes the types of the addresses where the collisions took place had 2011 missing values. I did not drop this column rather replaced the missing values with the label “others”, as it seems a potential candidate in determining the severity of an accident and hence could be a valuable input to our model. In the exact similar

way I also handled the missing values for the categorical features „JUNCTIONTYPE“, „WEATHER“, „ROADCOND“, „LIGHTCOND“ by replacing the missing values with the label “others” for the same reason.

The „INATTENTIONID“ feature which is a categorical variable and supposed to have a „Y“ for „yes“ and „N“ for „no“ values, but instead it had data inconsistency where the „yes“ values were labelled by „Y“ but the „no“ values were kept as missing values. I replaced all the missing values with 0's and replaced the „Y“ labels with 1's.

The „UNDERINFL“ feature had data inconsistency too. Some of the „yes“ values were labelled as „Y“ whereas the others as 1's and in a similar fashion some „no“ values were labelled as „N“ and the others as 0's. I handled the situation by converting all the 1's to „Y“s and all the 0's to N's. Moreover this feature also had 26479 missing values which I labelled as „UN“ which stands for „unknown“.

The „SPEEDING“ feature had data consistency where the „yes“ values were correctly labelled as „Y“s but the „no“ values were indicated using missing values. I replaced the missing values with 0's to indicate a „no“ and the „Y“s to 1“ to indicate a „yes“.

At last I checked whether the data-types of the features were consistent with the values they contained or not. I found out the „INATTENTIONID“ feature had a data-type inconsistency where the data-type of this feature was „object“ but the values it contained were 0's for a „no“ and 1's for a „yes“. Therefore, I converted the data-type of this feature from „object“ to „int64“ to make it consistent with the values it contained.